

WIEDERHOLUNG: KOMPLEXE ZAHLEN

Polarform und Inverse. Berechnen Sie Polarform und Inverse von:

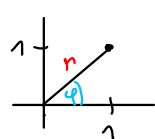
a) $z_1 = 1 + i$

b) z_1^4

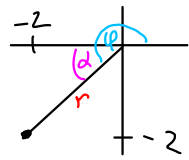
c) $z_3 = -2 - 2i$

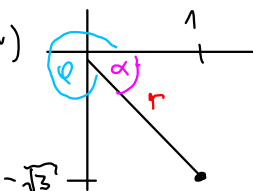
d) $z_2 = 1 - \sqrt{3}i$

Lösung.

a)  $r = \sqrt{1^2 + 1^2} = \sqrt{2}$
 $\varphi = \arctan\left(\frac{1}{1}\right) = 45^\circ \hat{=} \frac{\pi}{4}$ | $z_1 = \sqrt{2} e^{i\frac{\pi}{4}}, \quad z_1^{-1} = \frac{1}{\sqrt{2}} e^{-i\frac{\pi}{4}}$
ODER: $z_1^{-1} = \frac{1}{z_1} = \frac{1}{z_1 \bar{z}_1} \bar{z}_1 = \frac{1}{1^2 + 1^2} (1 - i) = \frac{1}{2} (1 - i)$

b) $z_1^4 = \left(\sqrt{2} e^{i\frac{\pi}{4}}\right)^4 = \sqrt{2}^4 \cdot e^{i\pi} = 4 \cdot e^{i\pi} \leftarrow \text{Polarform}$
ODER: $z_1^4 = (1+i)^4 = \underbrace{((1+i)^2)^2}_{1+2i+i^2 = -1} = (2i)^2 = 4i^2 = -4 = 4 \cdot e^{i\pi}$ | $(z_1^4)^{-1} = \frac{1}{-4} = -\frac{1}{4}$


c)  $r = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$
 $\varphi = \alpha + \pi = \arctan\left(\frac{2}{2}\right) + \pi = \frac{5\pi}{4}$ | $z_3 = 2\sqrt{2} e^{i\frac{5\pi}{4}}, \quad z_3^{-1} = \frac{1}{2\sqrt{2}} e^{-i\frac{5\pi}{4}}$

d)  $r = \sqrt{1^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$
 $\varphi = 2\pi - \alpha = 2\pi - \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{6\pi - \pi}{3} = \frac{5\pi}{3}$ | $z_2 = 2 \cdot e^{i\frac{5\pi}{3}}, \quad z_2^{-1} = \frac{1}{2} \cdot e^{-i\frac{5\pi}{3}}$

Eigener Lösungsversuch.

Wurzeln von komplexen Zahlen. Berechnen Sie die 4. Wurzeln von $1 + i$, d.h. alle z mit $z^4 = 1 + i$. Skizzieren Sie die Lösungen!

Lösung.

$$z^4 = 1 + i \stackrel{\text{s.o.}}{=} \sqrt{2} e^{i \frac{\pi}{4}} \quad \text{besitzt Lösungen:}$$

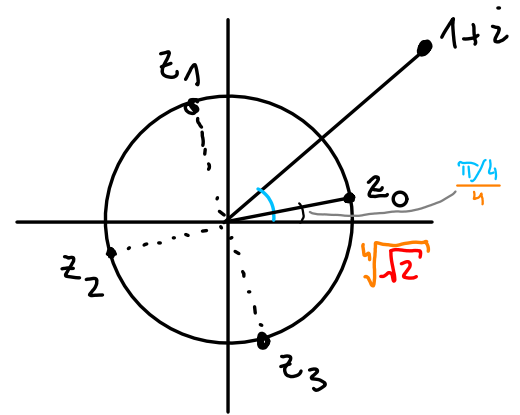
$$z_0 = \sqrt[4]{\sqrt{2}} e^{i \left(\frac{\pi/4}{4} + 0 \cdot \frac{2\pi}{4} \right)}$$

$$z_1 = \sqrt[4]{\sqrt{2}} e^{i \left(\frac{\pi/4}{4} + 1 \cdot \frac{2\pi}{4} \right)}$$

$$z_2 = \sqrt[4]{\sqrt{2}} e^{i \left(\frac{\pi/4}{4} + 2 \cdot \frac{2\pi}{4} \right)}$$

$$z_3 = \sqrt[4]{\sqrt{2}} e^{i \left(\frac{\pi/4}{4} + 3 \cdot \frac{2\pi}{4} \right)}$$

$$\underbrace{(2^{1/2})^{1/4}}_{= \sqrt[4]{2}}$$



ODER: $z_k = \sqrt[4]{\sqrt{2}} e^{i \left(\frac{\pi/4}{4} + k \frac{2\pi}{4} \right)} \quad \text{mit } k = 0, 1, 2, 3$

Eigener Lösungsversuch.